

Homework 6

Please give your final answer as a number where possible as well as show work for full credit.

Question 1. Please read chapter 9 of Chartrand et al. and write a couple sentences about a topic/example/concept that you found difficult or interesting and why?

Question 2. What is the exact term in $\left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^{10}$ containing x ?

$$\begin{aligned} \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^{10} &= \sum_{k=0}^{10} \binom{10}{k} (\sqrt{x})^{10-k} \left(\frac{-1}{\sqrt{x}}\right)^k \\ x^{(10-k)/2} (-x^{-k/2}) &= x^{\frac{10-k-k}{2}} (-1)^k = x^{5-k} (-1)^k \\ \sum_{k=0}^{10} \binom{10}{k} x^{5-k} (-1)^k & \\ \text{When } k=4, \binom{10}{4}, & \quad x^{5-4} = x^1 = x \end{aligned}$$

Question 3. Use the Binomial Theorem to simplify the sum,

$$\begin{aligned} &2^{n^2} + \binom{n}{1} 2^{n(n-1)} + \binom{n}{2} 2^{n(n-2)} + \cdots + \binom{n}{n-1} 2^n + 1 \\ &= \sum_{k=0}^n \binom{n}{k} (2^n)^{n-k} 1^k = (2^n + 1)^n \end{aligned}$$

Question 4. How many different 8-digit numbers can be obtained by permuting the digits in the number 70,440,704? (Since each number is an 8-digit number, the first digit cannot be 0.)

$$\frac{8!}{3!2!1!1!}$$

Question 5. A professor has 10 identical new pens that he no longer needs. In how many ways can these pens be given to 3 students if

- (a) there are no other conditions?
- (b) every student must receive at least one pen?
- (c) every student must receive at least two pens?
- (d) every student must receive at least three pens?

$$\begin{aligned} \text{(a)} \quad &\binom{10+3-1}{3-1} = \binom{12}{2} \\ \text{(b)} \quad &\binom{10-3+3-1}{3-1} = \binom{9}{2} \\ \text{(c)} \quad &\binom{10-6+3-1}{3-1} = \binom{6}{2} \\ \text{(d)} \quad &\binom{10-9+3-1}{3-1} = \binom{3}{2} \end{aligned}$$

Question 6. How many 4-digit numbers are there, the sum of whose digits is 11?

Question 7. A man enters Ben's Bagels to buy a dozen bagels only to learn that even though Ben has a large supply of plain bagels and blueberry bagels, he only has two cinnamon bagels left and no other kinds.

- (a) Express the number of selections of r bagels ($r \geq 0$) the man can make as a product of polynomials and/or power series¹.
- (b) Determine the coefficient of x^{12} of the product in (a). What information does this coefficient provide?

¹The book has a list of common power series in figure 9.8 on page 347.

Question 8. Suppose that we are interested in the number of non-negative integer solutions a, b and c of the equation $a + 2b + 3c = r$, where $r \in \mathbb{N}$.

- (a) What is this number when $r \in \{1, 2, 3, 4\}$?
- (b) Use generating functions to describe this number for a general $r \in \mathbb{N}$.

Question 9. A man buys 6 doughnuts, each of which is a plain doughnut, a powdered doughnut or a glazed doughnut. How many possible selections are there if he buys at least one doughnut of each kind?

- (a) Answer the question above by listing all possible selections in a table.
- (b) Answer the question above by determining the coefficient of x^6 in a product of polynomials and/or power series.
- (c) Answer the question above by computing $\binom{s+t-1}{s}$ for an appropriate choice of s and t .